

# CINERGY: Deterministic Power Monitoring for Carbon Accounting in the Cloud

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**Abstract**—Cloud providers now commonly offer tools to monitor the environmental impact of hosted services, aiming to inform customers about the carbon footprint of their infrastructures. A significant share of this impact arises from the power consumption of servers. To allocate server-related emissions, these methodologies typically rely on resource-quantity-based attribution and do not account for the effective usage of virtual resources. We demonstrate that resource usage can drastically affect carbon estimations.

This article proposes a framework, called CINERGY, to standardize how cloud providers integrate the power consumption of virtual resources into their carbon calculators. It defines a baseline to deterministically assess the power consumption of a provisioned *Virtual Machine* (VM)—i.e., the same usage should yield the same power estimate—while still accounting for consolidation gains enabled by virtualization. Our approach, constructed and evaluated empirically, achieves high precision, with a mean absolute error of 6.6% across various hardware.

Using metrics collected from cloud providers, we reveal the impact of instance type on energy efficiency, with differences reaching 85% for the same VM size. We also show that quantity-based carbon accounting methodologies can drastically underestimate actual CO<sub>2</sub> emissions—by up to a factor of three.

**Index Terms**—Cloud computing, Power model, Carbon Accounting

## I. INTRODUCTION

Efficiently reporting the environmental impact of cloud services is an ongoing effort within the community, notably driven by the rapid growth of cloud workloads. For example, between 2015 and 2022 (a period preceding the LLM era), cloud workloads increased by 350%, while data center energy consumption reached about 1–1.5% of global electricity use [2]. Thus, companies must monitor the environmental implications of their cloud usage. Cloud providers have developed methods and tools known as "carbon calculators," in which the impact (expressed as the eqCO<sub>2</sub> metric) of operated servers is allocated among customers [3], [4], [5].

However, even though reporting CO<sub>2</sub> indirect emissions is mandatory for large companies in some countries, there is still no clear methodology for reporting emissions associated with virtual resources. Cloud providers typically estimate their emissions following the *Greenhouse Gas* (GHG) Protocol [6],

but they are still lacking a standardized model [7]. This lack of standardization limits data comparability and obscures the metrics themselves—as many methodologies are not publicly disclosed—reducing their usefulness for emission-reduction efforts.

We argue that, beyond mandatory reporting, environmental impact reporting across different cloud services can serve as a powerful lever for cloud optimization. While the optimization of software power consumption is a well-studied field [8], the environmental impact of selected cloud products (e.g., instance types) remains underexplored.

This article contributes to more transparent carbon accounting methodologies by focusing on the evaluation of the power consumption of virtual resources, such as those offered through *Infrastructure-as-a-Service* (IAAS) and *Platform-as-a-Service* (PAAS) models. Depending on the carbon intensity of the electricity mix, operational power consumption can be the primary contributor to a server's CO<sub>2</sub> emissions [9], [10], outweighing manufacturing impacts. Notably, many cloud data centers are located in regions with high carbon intensity (e.g., the United States).

Accounting for the environmental impact of power consumption remains challenging. Cloud providers typically allocate estimated impact based on the amount of virtual resources requested by clients. Such static ratios may be sufficient for distributing hardware manufacturing impacts, but they fail to reflect the effects of actual usage (how a client consumes its resources) and platform load (which affects overall efficiency). This work shows that both factors can significantly alter power estimates and, consequently, the associated carbon emissions.

We introduce CINERGY, a framework for assessing the power consumption of virtual resources in operational conditions. Our contributions are summarized as follows:

- 1) we review existing work on power profiling software and its limitations for carbon accounting purposes;
- 2) we introduce a framework for evaluating the power consumption of *Virtual Machines* (VMs) while integrating load effects;
- 3) we quantify the energy efficiency across various cloud instance types, highlighting substantial disparities;
- 4) we show that providers' carbon accounting tools can significantly underestimate operational emissions for certain profiles.

Our work demonstrates that carbon accounting tools can account for usage while remaining deterministic. Maintaining consistent readings for identical actions is crucial for interpretability. In addition, we partnered with OVHcloud, one of the largest European cloud providers, to apply our methodology on production platforms. Our findings show that VM energy efficiency strongly depends on context. We demonstrate that the average energy efficiency of different cloud instance types varies by up to 85% for the same VM size under high consolidation (e.g., in oversubscribed contexts). Furthermore, we show that current carbon accounting tools significantly overestimate platform efficiency, leading to a substantial underestimation of actual emissions.

We begin by reviewing related work on software power monitoring in Section II, then present our framework in Section III. Section IV evaluates the precision of the models on benchmark workloads, while Section V examines their application in an operational context, highlighting efficiency variations and the limitations of current carbon tracking approaches. Finally, we conclude in Section VI.

## II. RELATED WORK

In this section, we overview the context of software power consumption measurements in a cloud context and highlight its limits for carbon accounting.

### A. Measuring server power consumption

Server power consumption has been widely studied over the past decades. It can be either directly measured or estimated. Early studies relied primarily on external wattmeters and instrumented hardware (e.g., modified power lines), enabling fine-grained insight down to component-level power draw [11]. However, such setups are complex to generalize in large-scale datacenters due to equipment and cost constraints. They are therefore mainly used as a first step to build models based on proxies that are correlated with power and more widely available (e.g., assembly instruction, hardware performance counters).

In recent years, access to power meters has been democratized. Sensors directly integrated by manufacturers into existing components emerged, such as *Smart Power Distribution Unit* (PDU) or *Baseboard Management Controller* (BMC) sensors, accessible via IPMI [12], are commonly used depending on availability [13]. The most widely adopted method for energy measurement is the *Running Average Power Limit* (RAPL) interface, introduced in [14], to capture the power consumption of Intel/AMD processors.

Intel's RAPL interface enables monitoring of CPU power consumption across several domains, depending on the processor generation. These domains include the entire package, CPU cores (PP0), non-core components, such as the integrated GPU (PP1), DRAM, and others. RAPL delivers accurate measurements, as its readings are based on integrated voltage regulators available since the Intel Haswell architecture. AMD processors also expose RAPL-like interfaces; however, early implementations were based on software models rather than direct hardware measurements.

### B. Power profiling software

Evaluating the energy usage of specific applications requires moving below the component-level granularity to determine what proportion of the system's power can be attributed to each software entity. Following the terminology introduced in [15], we distinguish between energy calculators, energy measurement software, and power profiling software.

Energy calculators rely on static computations to estimate energy consumption by requiring users to fill in forms [16], [17]. We exclude them from our scope, as our approach focuses on proactive monitoring of VMs to compute their dynamic consumption. Monitoring is also easier to integrate in practice, given that input parameters like usage ratios, often requested by calculators, are difficult to capture accurately. VM workloads are typically non-linear and exhibit bursty behavior.

Energy measurement software tracks energy consumption over a time window associated with the execution of a given software [18], [19]. They measure the total system energy between two timestamps and attribute it entirely to the software under study. As such, they are only suitable in single-tenant contexts, where no co-located workloads interfere. We exclude them from our scope due to the inherently multi-tenant nature of the cloud environment.

Power profiling software reports power consumption at the application level. These tools are the focus of this study, as VMs appear as processes from the host perspective. Two main categories can be distinguished: model-based and measure-based estimation.

Model-based profiling can be characterized as top-down: starting from activity-related metrics available at the process level (the top), it infers the underlying hardware power consumption. Models built on server component activity (e.g., CPU, memory, disk), such as those in [20], [21], can be applied to metrics collected from specific processes (e.g., VMs or containers). Alternatively, models can be trained directly on specific VM workloads, leveraging hypervisor-level metrics or more generic indicators (e.g., CPU usage) to accommodate workload diversity [11]. Model-based approaches provide determinism—identical usage yields identical estimates—but strongly depend on underlying assumptions, notably the choice of metrics, model structure, and operating conditions (e.g., with or without Dynamic Voltage and Frequency Scaling (DVFS), with or without co-located workloads), which may diverge from real operational contexts.

Measurement-based profiling leverages now-common internal sensors to retrieve power draw. This type of profiling can be described as bottom-up: starting from a platform-level power source (the bottom), they infer the contribution of individual processes (the top) using suitable proxies. CPU time is a common proxy and is used by monitoring tools, such as SCAPHANDRE [22] and JOULARJX [23]. Other tools, like KEPLER, rely on performance counters in bare-metal environments or leverage eBPF to extract metrics within VMs. Some approaches emphasize adaptability by dynamically selecting proxies at runtime. For example, SELFWATTS [24] (integrated into the POWERAPI framework [25], [26]) continuously iden-

tifies the performance counters most correlated with power consumption. Others, like KEPLER VM, may overlook cloud consolidation effects by relying solely on guest OS metrics.

While bottom-up approaches better reflect operational conditions by accounting for actual power draw, they face critical limitations for carbon accounting due to their non-deterministic nature. In particular, the energy efficiency of a computing platform is not constant and varies with load and consolidation. As emphasized in [27], power consumption is contextual: it depends not only on how much a process consumes but also on the system's overall load. For example, a process using 20% of the CPU contributes very differently to the server's total power depending on whether the server is idle or already at 80% utilization. In a cloud environment, this introduces a challenge: customers may receive different energy estimates for identical activities depending on the behavior of co-located tenants. Since customers have no control over server placement—an internal decision of the cloud provider [28]—this undermines consistency in accounting. Additional variability comes from tools that perform dynamic reconfiguration, such as SELFWATTS that adapt their models at runtime.

Several other power models specifically designed for cloud environments have been proposed, targeting both VMs [29], [30] and containers [31]. BITWATTS [32] distributes real-time power consumption across VMs using a model based on CPU frequency and unhalted cycles, but shares the limitations of process-level power models. JOULEMETER [29] constructs per-VM linear models based on CPU, memory, and disk usage, after a dedicated profiling phase. However, this approach becomes impractical at scale, as VMs often remain idle for long periods before becoming active. VMETER [30] similarly relies on linear models trained on performance counters, with workload-specific weights calibrated for each benchmark. In contrast, our objective is to develop a more generalized approach capable of handling diverse and previously unseen workloads effectively.

### C. Carbon tracking tools

Most cloud providers offer online tools to estimate the environmental impact of their services. These tools usually follow the GHG Protocol [6], which structures emissions into scopes. We focus on energy-related emissions falling under Scope 2—i.e., emissions associated with electricity production.

For example, Azure [4] estimates regional carbon emissions from data-center-level power measurements and allocates this impact to customers using a normalized cost metric, resulting in a static attribution for a given VM type. The Google Cloud Platform (GCP) [3] adopts a similar approach, allocating the impact of the platform based on normalized resources and similarly abstracting the actual client usage of resources. One distinction is that platform impact is not measured at the data-center level but at the equipment level: internal service power consumption is measured and then allocated to products relying on these services (proportionally to resource quantity or internal pricing) at the regional level, enabling final allocation to customers from aggregated measurements.

In the GHG Protocol [6] guidance, the approaches used by Azure and Google can be viewed as applications of the method termed *Single data point power consumption* (based on VM size and average power consumption) within the “virtual machine power assessment” category. However, this method is described as suitable only for static workloads, whereas cloud workloads are inherently highly dynamic.

OVHcloud follows a different approach to assess the environmental impact [33]. It attributes to each VM a share of the maximum power measured for the corresponding host type, proportional to the requested size, and then adjusts this value using the *Power Usage Efficiency* (PUE) and location-specific electricity emission factors. This approach is closer to GHG recommendations for dynamic workloads, as it accounts for host-level utilization (the method referred to as *Dynamic power consumption calculations*). However, it still does not account for the actual usage of virtual resources by clients.

Not all providers disclose their accounting methodologies [5], which limits comparability and motivates the need for a common framework. This lack of transparency reinforces the need for a unified and robust approach to virtualized power accounting.

One reason current approaches often ignore usage is the non-determinism it introduces due to workload interference and load-dependent energy efficiency. This work contributes to improving the GHG assessment of VM power consumption using hypervisor-level information.

### D. Synthesis

We propose to integrate customers' virtual power usage and cloud providers' virtualization efficiency into carbon accounting mechanisms. To the best of our knowledge, this article is the first to assess the power consumption of virtual resources in real cloud environment conditions, where both the tenant usage and the platform load play a significant role. While our framework may leverage the same metrics used in existing power profiling software, our objective differs, as we estimate a virtual representative (yet deterministic) power consumption.

## III. CINERGY FRAMEWORK

We now present the core principles of our methodology. Our approach combines the strengths of top-down power profiling (ensuring determinism) with those of bottom-up profiling (capturing the representativeness of runtime conditions).

To obtain a non-contextual estimate of a VM's power consumption, the framework first models its behavior under a reference scenario. This baseline scenario, referred to as the Worst Case Power Consumption (WCPC), is detailed in Subsection III-A along with guidelines for its construction.

Next, to account for the energy savings enabled by virtualization and workload consolidation, the framework adjusts this baseline using a per-instance-type energy efficiency model. The methodology for modeling this efficiency and applying it to the WCPC is described in Subsection III-B.



### A. Worst Case Power Consumption

In this subsection, we describe how to model the base power consumption used in the CINERGY framework. Specifically, we aim to estimate the power consumption of a VM running in isolation—that is, on a dedicated host—where all system power can be attributed to the single workload. This situation is referred to as the *worst case*, as servers are typically more energy efficient at higher usage [34].

To address this, we identify three key research questions:

- 1) How can WCPC be estimated?
- 2) How should static power consumption be distributed?
- 3) How can WCPC be applied in real-world scenarios?

We answer these questions in the following subsections and discuss the design choices made in the framework.

1) *Estimating the WCPC*: To ensure determinism in our power estimates, we adopt an approach inspired by top-down power profiling. We rely on metrics collected during process execution and infer the power that would be consumed by the underlying system. As discussed earlier, such approaches depend on modeling assumptions; here, we explicitly target a scenario in which the VM runs in isolation. Unlike existing methods that derive VM-level estimates from generic server models, we build models directly from measurements on systems hosting an isolated VM, ensuring results that accurately reflect its behavior.

Measuring the software’s power consumption in isolation reduces two primary sources of bias. The first is interference from noisy neighbors—co-running threads that affect system power consumption. The second is kernel activity variability: by fixing the system configuration (hardware and OS), we create a consistent baseline. Our goal is to model power in this default setting. We now examine how the model must be tailored to the specific context of our framework.

#### 2) Modeling power consumption:

a) *Load range*: As cloud VMs typically have a small number of *virtual CPUs* (vCPUs) relative to their host servers [35], it is essential to model their isolated power consumption on larger machines, particularly in low-usage regimes. Interestingly, the low-usage modeling of servers has certain limitations when based on overall data, an approach traditionally used by power profiling tools. Figure 1 illustrates this effect. The measurements obtained from the different CPU usage levels using RAPL were used to capture a third-order polynomial regression. This order was determined to be the most effective within the explored range. The regression performed well across the server range, achieving a coefficient of determination ( $R^2$ ) of 0.98 (with 1 being the maximum) and a mean absolute error of 1.7 W. However, this performance significantly decreases when focusing on the low-usage phase (below 15% CPU activity). In this low-usage phase, the mean absolute error nearly triples within the measures, increasing from 2.7 W to 4.6 W. This error is proportionally larger than the actual power values, as physical measurements are lower. Additionally, the coefficient of determination drops to 0.84 in this range. While the degree of degradation varies depending on the processor being tested, this issue consistently appeared on our testbed. The complexity of patterns during low-usage

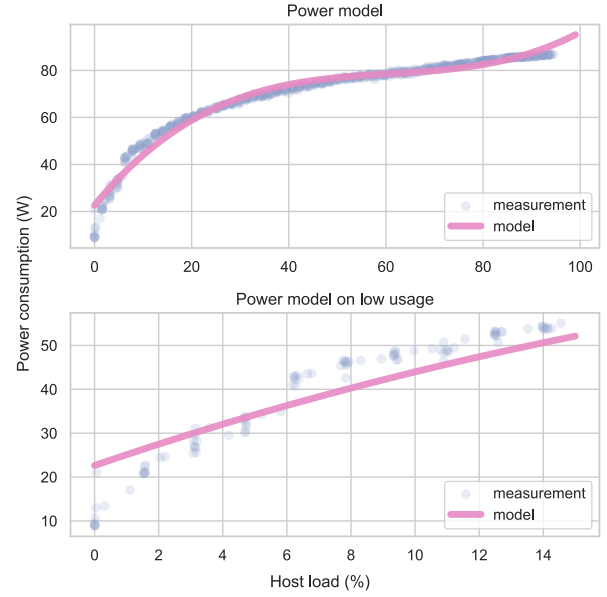


Fig. 1. Illustration of reduced precision in regular models for low usage on a Broadwell architecture (Xeon E5-2620 v4)

phases typically requires a more intricate modeling approach; for most tested processors, a polynomial regression of the fifth order was optimal to represent this specific range accurately.

WCPC models must target the relevant usage range to maintain precision. Specifically, when modeling the usage of a VM of  $n$  cores, we select the range as follows:  $[0; \frac{n}{\text{host cores}} \times \alpha]$ ,  $\alpha$  being a safety factor to ensure an accurate trend in the curve. In this study, we determined  $\alpha$  based on the number of cores of physical servers, aiming to limit the range to low CPU usage phases (up to  $\sim 40\%$ ). For instance,  $\alpha = 3$  is appropriate for VMs with 4 cores on hosts with 32 cores.

It should be noted that models are constructed from the host’s perspective: physical power readings (e.g., RAPL) are collected and correlated with host-level proxy metrics (e.g., CPU load). Adopting this perspective is essential to apply the models effectively. Consequently, VM usage must be translated into equivalent host-level metrics. For example, if a VM utilizes 100% of its 4 vCPUs, this should be represented as  $(\frac{4 \times 100}{m})\%$  of host CPU usage, where  $m$  is the total number of physical cores. Similar conversions must be applied when using other kinds of resources (e.g., memory, disks).

b) *Workloads*: The input workloads used to generate the power model should be carefully selected. Cloud workloads are inherently diverse and treated as black boxes. However, a common element across all VMs is the presence of a guest operating system and, by extension, a guest scheduler. In both bare-metal and virtualized environments, schedulers are responsible for distributing workloads across available computing resources to maximize performance. This strategy gradually loads shared resources—such as sibling cores and cache levels—rather than saturating specific ones, directly affecting power consumption through the host’s DVFS policy, which should remain configured as in production.

In practice, it is uncommon for all vCPUs to be fully utilized

simultaneously [36]. The guest kernel’s workload distribution depends on its parallelism. Consequently, representative workloads should be selected to capture power variations induced by DVFS. To explore this issue, we trained different models under varying core loads. Specifically, we considered synthetic benchmarks that stress 25%, 50%, and 100% of their thread time, until the host was fully loaded. Loading a host with processes that use 25% of their thread time, compared to those that use 100%, allows more processes to run while keeping overall CPU usage the same, leading to a more distributed workload. We empirically determined the most representative load level for a thread (analogous to a vCPU) among the 3 tested thread times by evaluating the accuracy of our trained models. We found that the 50% load level was the most representative.

3) *Distributing static power consumption*: Powering up any server to execute its operating system and a hypervisor leads to non-negligible static power consumption, notably because the power drawn by some server components, such as the motherboard, is not bound to resource usage. In servers without GPUs, the CPU is the primary power-consuming component; we propose allocating the static power proportionally according to its adjusted allocation. We, therefore, adopt a methodology similar to the approach described in [6] and attribute the static power consumption using Equation 1.

$$\mathcal{P}_{VM}^{static} = \frac{allocation}{host\ cores} \times \mathcal{P}_{server}^{static} \quad (1)$$

This results in a *de facto* power attribution to VMs that is not directly tied to their dynamic usage, as providers allocate resources even when they are not actively used. However, the allocated resources may differ from the actual number of vCPUs in oversubscribed contexts (also known as *overcommitment*), where virtual resources exceed the available physical resources on a server [37]. For static oversubscription—where the same ratio is applied uniformly across all servers—the number of cores per VM in Equation 1 should reflect the effective physical allocation. In contrast, dynamic oversubscription adjusts the oversubscription level based on each server’s actual usage rather than relying solely on the initial allocation [38], [39], [40]. In that specific case, we propose adjusting the VM allocation by dividing it by the average oversubscription ratio observed across the cluster.

4) *Applying to real-world systems*: We now discuss how WCPC estimations can be integrated into real-world systems, like production cloud platforms. To this end, we developed tools to support the implementation of the framework. While our methodology includes techniques from traditional power profiling tools, its purpose diverges, necessitating distinct design choices for both the *training* and *production* phases.

As shown in Figure 2, the framework starts with a training phase, where diverse workloads are used to calibrate the WCPC model for the targeted *Physical Machine* (PM) type. As a top-down power estimation approach, the resulting model is deterministic by construction. However, to better reflect operational conditions (similarly to bottom-up models), we apply a constant correction ratio, described in the next subsection (cf. III-B).

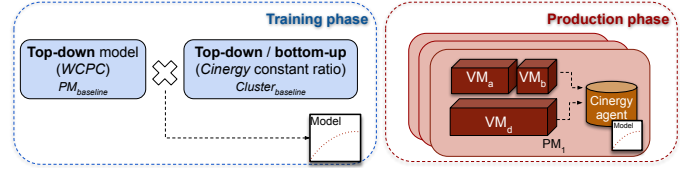


Fig. 2. Overview of the CINERGY power modeling framework

Once trained, the model is deployed on each physical host through a local agent that continuously monitors VM usage metrics. VMs can then query on-demand estimates of their power consumption.

We developed a Python toolkit (approximately 600 lines of code) to automate WCPC modeling. It generates diverse workloads, including training workloads, isolated VMs, and multiple concurrent VMs, and streamlines RAPL data collection across heterogeneous hardware. This tooling enables validation of model accuracy against ground-truth measurements.

The CINERGY agent exposing WCPC estimations to users is implemented in Rust, in approximately 250 lines of code. The underlying design philosophy emphasizes simplicity: it loads the trained model and applies it on demand to produce power estimates. To do so, it monitors system PIDs and identifies QEMU/KVM processes. For each detected VM, it tracks CPU usage—along with other potential proxies—via the `/proc` file system. VMs can query their current power usage through a REST endpoint exposed by the host.

### B. Cloud Instance Type Energy Efficiency

The WCPC represents a worst-case estimate, rarely reached in practice, as servers typically host multiple VMs simultaneously. This sharing increases resource utilization and improves energy efficiency, due to the sublinear (often logarithmic) shape of server power curves. Our goal is to capture these energy savings from workload consolidation and integrate them into our estimations.

In other words, while top-down power profiling provides deterministic usage-based estimates (WCPC), we also aim to account for actual runtime conditions captured by bottom-up power profiling tools.

$$\text{CINERGY ratio}_{VM} = \frac{\text{bottom-up model}_{VM}}{\text{top-down model}_{VM}} \quad (2)$$

To bridge this gap, we introduce a consolidation factor called the CINERGY ratio<sup>1</sup>. It is defined as the ratio between the runtime (bottom-up) power estimation and the isolated (top-down) power estimation for a given VM. Since isolated consumption is usually higher, the ratio typically falls between 0 and 1. For cloud customers, the CINERGY ratio reflects the energy efficiency gains enabled by infrastructure sharing. For example, a ratio of 0.4 indicates that only 40% of the power expected in isolation is effectively consumed—implying a 60% saving thanks to consolidation.

<sup>1</sup>Derived from “Cirrus,” a type of cloud, and “Synergy,” denoting an interaction producing a whole greater than the sum of its parts.

TABLE I  
PROCESSORS USED IN OUR EXPERIMENTS

| Vendor | Architecture    | CPU model                         | Cores / Threads | TDP  | Release year |
|--------|-----------------|-----------------------------------|-----------------|------|--------------|
| Intel  | Broadwell       | Xeon E5-2620 v4 ( $\times 2$ )    | 16 / 32         | 85W  | 2016         |
| Intel  | Cascade Lake-SP | Xeon Gold 5218R ( $\times 2$ )    | 40 / 80         | 125W | 2020         |
| Intel  | Ice Lake-SP     | Xeon Platinum 8358 ( $\times 2$ ) | 64 / 128        | 250W | 2021         |
| AMD    | Zen1            | EPYC 7351 ( $\times 2$ )          | 48 / 96         | 170W | 2017         |
| AMD    | Zen2            | EPYC 7352 ( $\times 2$ )          | 48 / 96         | 155W | 2019         |
| AMD    | Zen3            | EPYC 7513 ( $\times 1$ )          | 32 / 64         | 200W | 2021         |

Many power profiling tools from the literature [22], [41], [42] can be used as the bottom-up component. A key consideration is ensuring that the static allocation mechanism is consistent between the two models to give representative values.

In practice, we propose computing the CINERGY ratio per cluster, as each cloud instance type typically maps to a specific set of servers operated under similar constraints. Premium offerings with dedicated resources exhibit lower utilization due to the gap between provisioned and used resources. Conversely, oversubscription increases hardware utilization for instances with lower availability requirements.

The mean CINERGY ratio for a given instance type can then adjust the WCPC to reflect more realistic consumption while preserving determinism for carbon accounting. The CINERGY ratio notably promotes the savings resulting from server consolidation that keep being seen negatively despite its potential to make cloud computing greener [36], [40], [43].

#### IV. EMPIRICAL EVALUATION

In this section, we focus on evaluating how accurately the WCPC can be inferred from runtime conditions. Note that the CINERGY framework does not impose restrictions on the choice of power model or hardware to model (we discussed in the previous section the interest of using generic power models). However, this section demonstrates that top-down models can be accurate on representative hardware (Xeon and EPYC architectures), even when using a simple proxy.

To this end, we used the Python toolkit introduced earlier to evaluate CINERGY in three steps. First, we developed power models in a controlled environment for each hardware type (cf. Subsection IV-A). Next, we measured the activity and corresponding power consumption of a VM on a dedicated server to establish a ground-truth reference (cf. Subsection IV-B). Finally, we assessed the model's ability to replicate this ground truth in a cloud-like scenario, where the same VM ran alongside other processes on the host (cf. Subsection IV-C).

##### A. Training WCPC Power Models

Cloud data centers operate various processors from different vendors and generations, each possessing unique energy characteristics. To assess the broad applicability of our approach, we evaluated it on multiple processors, as detailed in Table I, paying careful attention to covering a wide diversity of architectures.

We trained power models for each server in a controlled environment using the CPU time as a proxy. Workloads

of varying intensity levels were created using `stress-ng`, a widely-used benchmarking tool<sup>2</sup> with its default stressor settings. We then gradually increased the host load by running threads that used 50% of a CPU core, until reaching its total capacity.

The collected energy consumption data includes RAPL measurements, including power usage for each CPU package. We aggregated the reported readings to compute the total power consumption of multi-socket servers. We then applied a polynomial regression of order 5 to convert the CPU load into its corresponding power consumption (in Watts). Note that while we chose to first evaluate the accuracy of simple models for the WCPC, our framework does not impose any restrictions on alternative modeling strategies.

##### B. Ground Truth Measurements

We launch a VM that runs a specific application under various CPU usage levels while monitoring the power consumption measured by RAPL domains. This setup serves as the baseline since we can fully attribute the power consumption recorded by RAPL to the system's single VM.

Our evaluation was executed on a VM with four vCPUs, while running the social network application from the Death-StarBench (DSB) [44]. The DSB features a microservices-based architecture, incorporating various components such as load balancers, proxies, message queues, databases, caches, and services. We chose this architecture because of its complexity and diverse components, which make it a strong candidate to benefit from energy monitoring capabilities. Additionally, the DSB includes a workload generator based on `wrk2`, enabling us to simulate realistic usage patterns during our evaluation.

The VM's activity was driven by the `wrk2` workload generator, simulating several load phases: *low* (100 rps), *medium* (2,000 rps), *high* (5,000 rps), and *very high* (10,000 rps). We selected these load levels based on the CPU utilization they generated. Under the low-load scenario, the VM was primarily idle, while the highest load completely engaged the CPU.

The CPU load of the VM and the corresponding CPU activity observed from the host were continuously monitored over several hours to ensure accurate and representative readings.

##### C. Models Evaluation in a Cloud-like Environment

Measurements of VM activity were conducted in a cloud-like environment, where a VM ran alongside additional pro-

<sup>2</sup><https://github.com/ColinIanKing/stress-ng>

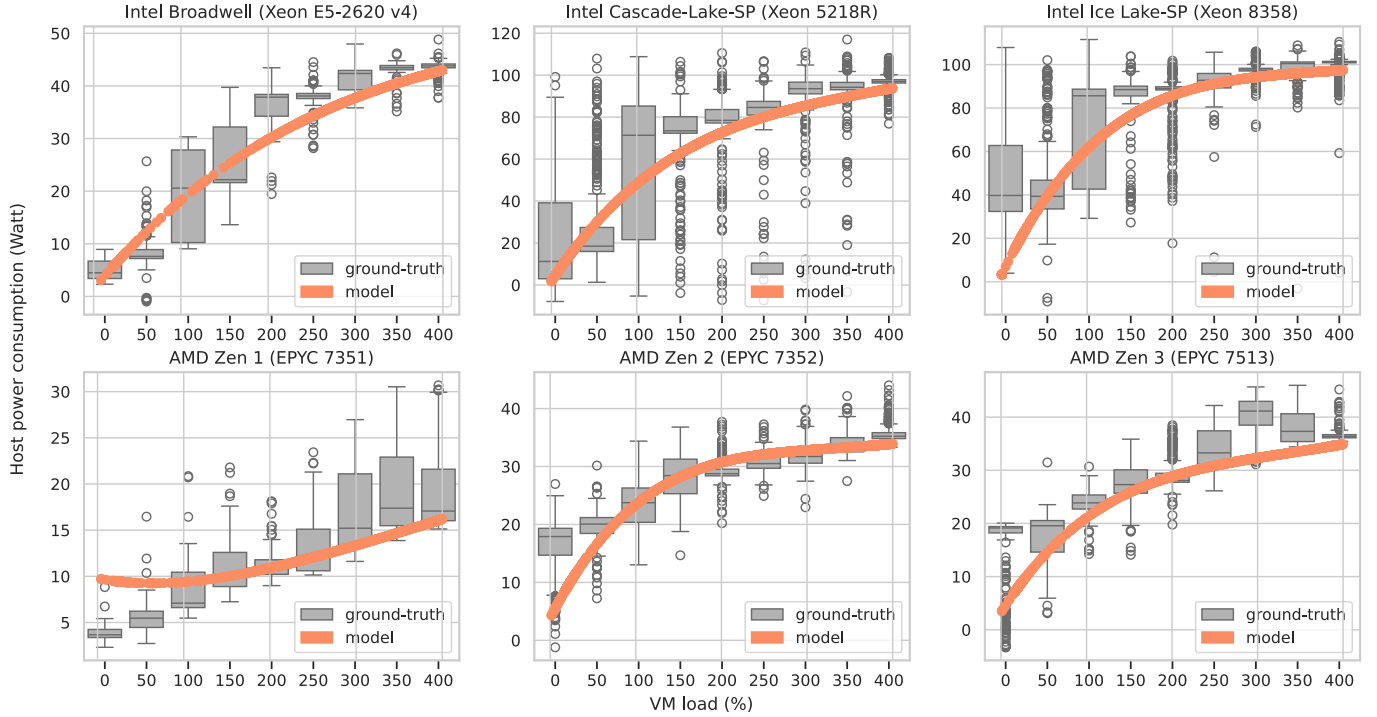


Fig. 3. WPCP power models trained with CINERGY for various server architectures

cesses to simulate background noise. The same VM workload, generated using `wrk2` under identical parameters, was repeated during these tests. Throughout the experiments, the VM activity was continuously monitored while background processes were gradually introduced, varying the host's load from mostly idle to fully saturated.

To assess the accuracy of our model under different conditions, we applied various load patterns (*low*, *medium*, *high*, and *very high*) to the VM in a cyclical sequence. This approach enabled us to evaluate the model's power estimations across a range of VM workloads and host load combinations.

Figure 3 depicts our WPCP power models across the studied architectures. The ground-truth measurements, discussed in Subsection IV-B, are depicted in grey, while our power model's predictions, which were trained using the generated data discussed in Subsection IV-A and the metrics collected from this cloud-like environment, are plotted in orange. While the VM activity leads to different host CPU usage depending on its size, results are displayed based on VM CPU activity (x-axis) to normalize the data. The static power consumption was allocated based on resource distribution in a non-oversubscribed setting for the ground truth and the power model predictions.

Despite some inherent variability in power consumption measurements [45], our power models report high accuracy in predicting power usage across different architectures. Below are the prediction statistics for WPCP estimations:

- Intel Xeon E5-2620 v4 (Broadwell, 85 W TDP): the model achieved a *Root Mean Squared Error* (RMSE) of 3.8 W and a *Mean Absolute Percentage Error* (MAPE) of 7.2%.
- Intel Xeon Gold 5218R (Cascade Lake-SP, 125 W TDP):

higher TDP and more threads resulted in a larger RMSE of 20.2 W and a MAPE of 15.1%.

- Intel Xeon Platinum 8358 (Ice Lake-SP, 250 W TDP): this high-TDP processor had an RMSE of 14.9 W and a MAPE of 5.5%.
- AMD EPYC 7351 (Zen1, 170 W TDP): prediction accuracy was high, despite a mostly linear pattern, with an RMSE of 6.8 W and a very low MAPE of 3.0%.
- AMD EPYC 7352 (Zen2, 155 W TDP): results showed an RMSE of 4.2 W and a MAPE of 2.6%.
- AMD EPYC 7513 (Zen3, 200 W TDP): this model reached an RMSE of 6.2 W and a MAPE of 6.0%.

Our findings suggest that the higher RMSE observed in some settings is largely influenced by the variability in the initial energy readings [45]. This behavior is inherent to fine-grained ground-truth measurements, where repeated executions of identical workloads can yield different power values due to hardware and system-level effects. Our approach enforces deterministic power attribution, ensuring that identical VM activities are mapped to identical power estimates, without introducing systematic bias. While this comparison against noisy reference measurements inflates error metrics, it does not affect order-of-magnitude accuracy, which is sufficient for carbon accounting.

AMD's Zen processors showed lower MAPE, likely because their RAPL energy counters provide more stable readings that simplify modeling compared with noisier measurements. Overall, CINERGY models perform well across all tested architectures. The average MAPE for all evaluated processors is approximately 6.6%, indicating a low mean relative error



in the predictions. This supports the feasibility of accurate isolated power estimation in diverse hardware environments.

#### D. Discussion & Limitations

We evaluated WCPC accuracy using a model based solely on CPU time and RAPL measurements, as these are commonly used by existing power profiling tools. While this approach achieves good accuracy on the evaluated generic workloads, it has important limitations. In particular, identical CPU time values can correspond to different power consumptions depending on the instruction mix and memory behavior (e.g., scalar versus vector units, cache intensity), which can bias estimates for specific workloads. In addition, RAPL captures only socket-level components (cores, uncore, cache) and excludes other resources such as network and storage, leading to an underestimation of total consumption. More fine-grained models incorporating additional performance counters or resource-specific measurements could therefore improve WCPC accuracy for heterogeneous workloads.

Our evaluation used production-class servers to ensure a measurable baseline for model accuracy, as isolated software power consumption can be directly measured—unlike bottom-up methods that infer it indirectly. This setup emphasizes higher power values due to the servers' static consumption. Choosing an appropriate server baseline is context-specific. Instead, our framework focuses on quantifying energy efficiency savings from large-scale consolidation. To that end, we compute a VM's worst-case consumption and apply the average CINERGY ratio of the cluster, yielding a deterministic yet realistic estimate that accounts for typical co-hosting. The next section provides examples of how the ratio behaves in a cloud-like setting.

#### V. APPLYING THE CINERGY RATIO

This section focuses on interpreting the CINERGY ratio in a cloud computing context. In Subsection V-A, we provide an overview of consolidation gains for different VM profiles. Then, in Subsection V-B, we present insights from a partnership with OVHcloud—one of the largest European cloud providers—on how energy efficiency varies across cloud offers. Finally, Subsection V-C applies the CINERGY framework to carbon accounting estimates, highlighting differences compared to current methodologies.

##### A. Consolidation efficiency across VM profiles

To illustrate the effect of the CINERGY ratio on different VM profiles, we conducted an experiment where a VM power consumption was measured under different host CPU loads. We computed power consumption estimates from our top-down model and a bottom-up-like model using Equation 3.

$$P_{VM}^{\text{bottom-up}} = \frac{CPU_{VM}^{usage}}{CPU_{total}^{usage}} \times P_{RAPL} \quad (3)$$

Figure 4 illustrates the power estimation obtained from both top-down and bottom-up models when applied on the 4-vCPU VM. We distinguished two VM usage phases: *low*

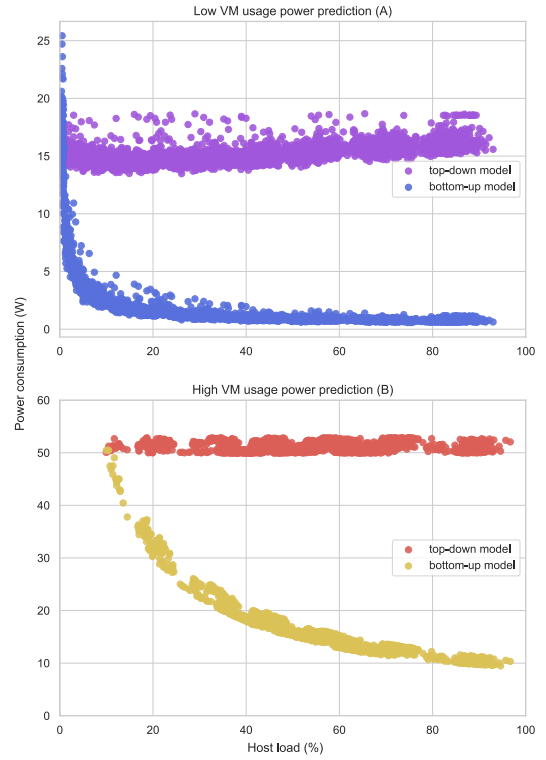


Fig. 4. Power prediction for two types of VMs of bottom-up and top-down models

*usage* (below 10%, subplot A) and *high usage* (above 85%, subplot B).

The estimation of our top-down model was steady on both low-usage and high-usage phases (around 17 W and 50 W, respectively). Note that the slight variations observed in the figures do not come from non-determinism but from the range of CPU loads (for CPU load qualified as low, there is a power reading gap between 0% and 10% CPU load).

In contrast, the power estimation reported by the bottom-up approach varied substantially. For example, for a VM reporting high CPU usage (subplot B), the bottom-up model estimates the same 50 W when the host load is low (as expected, because it is similar to an isolated context) but decreases to up to 10 W with a fully loaded host. It represents an 80% variation in the power estimation due to external VM activities.

With VMs under low load (subplot A), the two models diverge, even when host activity is minimal. In such scenarios, the bottom-up model, which relies on live RAPL measurements (as described in Equation 3), can occasionally report higher values than our isolated estimation. These anomalies stem from the variability inherent in power readings during low host load, where any kernel activity causing power spikes is fully attributed to the VM, resulting in random peaks. Our top-down approach mitigates this variability, highlighting its suitability even in predominantly dedicated host scenarios. Interestingly, under low usage, the bottom-up model's power estimation tends toward zero as the VM's idle resources are progressively used by other processes, mimicking behavior observed in dynamically oversubscribed environments.

Subsequently, we analyzed the CINERGY ratio variation



for this VM as other processes introduced additional load in Figure 5. It summarizes the two scenarios by illustrating the evolution of the CINERGY ratio as a function of host load. To emphasize the energy efficiency gains associated with increased host utilization, CINERGY values exceeding one—caused by outlier readings in the bottom-up model—are not displayed. The results show that energy efficiency improvements become evident even at relatively low levels of host utilization. At a 20% host load, the CINERGY ratio demonstrates significant energy efficiency savings for both VMs. Specifically, the high-usage VM operates at 60% of its baseline power consumption ( $r = 0.6$ ), while the low-usage VM sees even greater benefits, consuming just 10% ( $r = 0.1$ ) of its initial power. As the host load approaches full capacity, the high-usage VM's consumption decreases further to 20% of its isolated power consumption ( $r = 0.2$ ), while the low-usage VM trends towards 0.

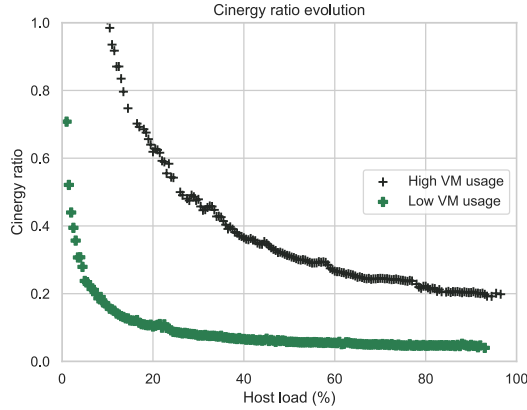


Fig. 5. CINERGY ratio evolution for different VM profiles depending on the host load

### B. Efficiency Variations Across Instance Types

Cloud providers offer different products, varying the managed stack (from IAAS to *Software as a Service* (SaaS)) and the availability level. We now focus on illustrating energy efficiency variations between two cloud instance types. To that end, we partnered with OVHcloud. They notably propose different VM-based products. One, referred to as *Public Cloud*, guarantees resources—and therefore performance—to customers. Another, referred to as *Virtual Private Server* (VPS), offers lower guarantees and shares physical resources among tenants.

We retrieved traces from two distinct clusters of servers, each operating one of the instance types. Both clusters comprise more than 1,000 physical servers and were studied over one week. We collected their CPU load every 5 minutes and converted it into a CINERGY ratio to better assess their energy efficiency. Specifically, we used the CINERGY ratio model from Figure 5 for the high VM profile.

Distribution of ratios is reported in Figure 6 as a *Kernel Density Estimate* (KDE). Unsurprisingly, VPS hosts reached higher load due to their oversubscribed context, and therefore better energy efficiency. While some *Public Cloud* hosts

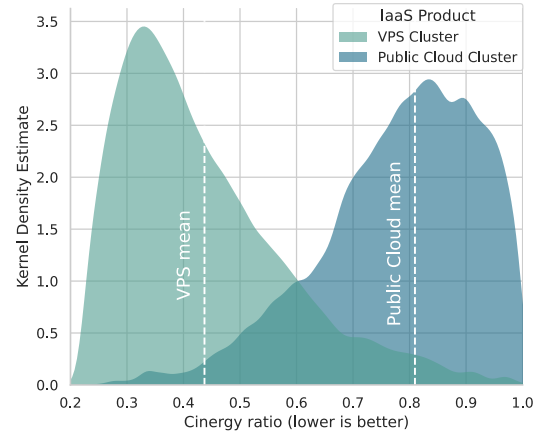


Fig. 6. Distribution of CINERGY ratio obtained by hosts of two different OVHcloud instance types over a week

achieve very high usage—and thus high energy efficiency with a ratio that can fall below 0.3—this remains an exception, as all provisioned resources must be fully used simultaneously to reach this level. In practice, the average CINERGY ratio is 0.81 for *Public Cloud* VMs, representing only a 19% power reduction compared to the worst case. On the other hand, *VPS* VMs have an average ratio of 0.44 (56% power reduction). Comparing both ratios highlights that *VPS* VMs are approximately  $1.85\times$  more energy efficient than *Public Cloud* ones, on average.

This illustrates the trade-off between energy efficiency and high availability—a choice that customers must consider according to their needs. We believe that the CINERGY ratio is an insightful metric to assist customers in capturing this trade-off. The average CINERGY ratio per cluster (or instance type) can be published to preserve the determinism of our method when applied to worst-case power estimates.

In the studied dataset, variations in the CINERGY ratio across hosts can only be partially explained by allocation percentage. Hosts considered fully allocated exhibit only a marginally improved CINERGY ratio compared to the cluster average (e.g., 0.40 vs. 0.44 for *VPS*). This reflects a provider-side consolidation strategy in which provisioned capacity does not excessively exceed demand, improving overall cluster efficiency and, consequently, the CINERGY metric.

Workload size also affects the metric. On hosts considered fully allocated, we observe a correlation between the number of instances and the achieved efficiency. Hosts filled with many small VMs tend to be more heavily utilized and therefore exhibit a lower CINERGY ratio than hosts allocated to fewer, larger VMs. This trend holds for both *VPS* and *Public Cloud* products, highlighting that small VMs sizes play an important role in cloud catalogs, not only for packing efficiency but also for achieving higher server utilization.

Note that, while differences between clusters here stemmed from oversubscription, additional mechanisms such as spot VMs and harvesting VMs exist among cloud providers [46], [47] to improve energy efficiency. These would also enhance the CINERGY ratio.

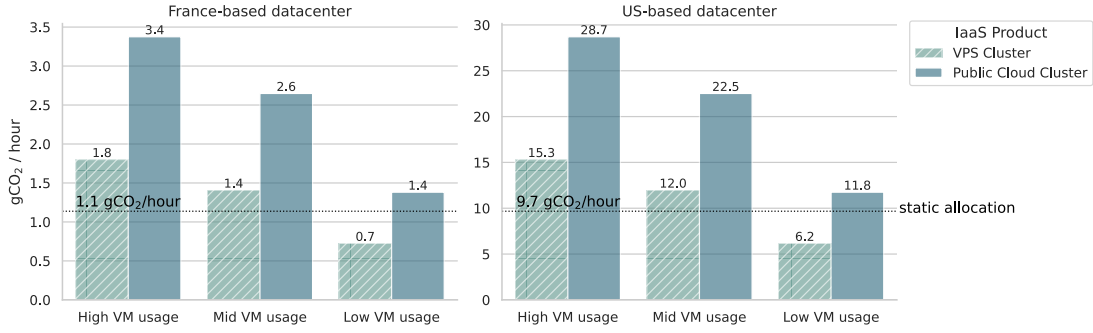


Fig. 7. Scope 2 carbon estimation (gCO<sub>2</sub>) across datacenters, VM usage profiles (4 cores), and instance types. Measurements are based on a Xeon E5-2620 v4 ×2 server, using an external wattmeter to measure maximum and minimum power for static-allocation-based estimations.

### C. Impact on Carbon Accounting Tools

Finally, we highlight the impact of our power estimation method on carbon accounting tools. We focus on Scope 2 of carbon accounting methodologies, which reports indirect emissions associated with electricity consumption, notably from servers. We use the current OVHcloud carbon accounting methodology [33] (as of early 2026) as a baseline for several reasons: the CINERGY ratio was evaluated in this context; the methodology and required data are publicly available; and it relies on server-level measurements, unlike other providers' approaches that use more aggregated attributions.

$$\text{footprint} = \text{power estimate} \times \text{PUE} \times \text{emission factor} \quad (4)$$

As defined in Equation 4, the usage footprint is computed from the power estimate, the data center PUE, and a location-based emission factor sourced from the ElectricityMap database. Note that the product of PUE and the emission factor approximates the Carbon Usage Effectiveness (CUE) metric [48].

In the OVHcloud methodology, the power estimate corresponds to a fraction of the server's measured maximum power consumption, obtained using an external wattmeter under full load and allocated proportionally to the VM size. In our methodology, the static power reference is likewise measured using an external wattmeter and dispatched in the same manner (as described in Equation 1), while dynamic power consumption, derived from RAPL, is estimated as the WCPC of the VM multiplied by the CINERGY ratio.

Figure 7 reports different Scope 2 carbon footprint estimations. Two OVHcloud datacenter contexts are shown: a datacenter located in France (Gravelines, PUE 1.22, electricity mix 56 gCO<sub>2</sub>/kWh) and a datacenter located in the United States (Vint Hill, Virginia, PUE 1.39, electricity mix 418 gCO<sub>2</sub>/kWh).

As our method accounts for both actual VM usage (via the WCPC) and the consolidation efficiency of the instance type (via the CINERGY ratio), we consider a 4-core example VM deployed on a representative host (Xeon E5-2620 v4 ×2) and apply the analysis to the two cloud products previously evaluated: *VPS* and *Public Cloud*. Three VM usage profiles are studied: *low usage* (below 10%), *mid usage* (45–55%), and

*high usage* (above 85%). The horizontal dotted line indicates the current estimation for a 1-hour activity of this 4-core VM, which is static for a given datacenter, relying solely on size-based attribution.

For moderately to highly utilized resources, the current static estimation underestimates CO<sub>2</sub> emissions in both datacenters. This occurs because proportional attribution based on maximum power implicitly assumes a linear power model and, therefore, a constant efficiency. However, servers are less efficient at low-to-mid load (the typical operating range in cloud environments) than at full load, following an approximately logarithmic power-load relationship. As a result, linear allocation assigns less power than what is observed under operational conditions. The effect is more pronounced for *Public Cloud* products due to their lower average server utilization (i.e., lower CINERGY ratio).

The gap can be substantial, especially when extrapolated to monthly emissions in high-carbon-intensity contexts (higher PUE and higher electricity carbon intensity), as observed in the US-based example. When scaling hourly emissions to one month, our estimation reaches 11.0 kgCO<sub>2</sub> for an actively used 4-core *VPS* VM (4.0 kgCO<sub>2</sub> more than the linear allocation), and 20.7 kgCO<sub>2</sub> for the equivalent *Public Cloud* VM, corresponding to 3.0× the current static estimate. Methodologies used by other providers that rely on aggregated values (DC-scope or service-scope) are also likely biased in estimating individual efficiency, since cluster characteristics are often heterogeneous.

Conversely, the CO<sub>2</sub> emission estimate can be similar or even lower for low or idle VM usage, especially in over-subscribed contexts. As utilization decreases, the estimation primarily assigns to the VM the emissions corresponding to static power components, which can yield values lower than those obtained through proportional allocation based on peak power. For *VPS*, we assume an oversubscription ratio of 2, which further reduces the attributed share. This illustrates that oversubscription, while improving aggregate energy efficiency, also enables providers to amortize unused capacity. From a user perspective, emissions can therefore be reduced by removing unused VMs, resizing underutilized instances, or selecting oversubscribed offerings for intermittent workloads.

Overall, this justifies integrating resource usage — both at the VM level and at the cluster level — into carbon accounting

tools to improve representativity and support more effective efficiency-oriented policies.

## VI. CONCLUSION

This article addressed the problem of power monitoring *Virtual Machines* (VMs) for carbon accounting needs. We propose a framework to integrate virtual deterministic power consumption into carbon accounting methodologies. It relies on modeling the Worst Case Power Consumption (WCPC) of the VM, and applying a factor capturing the consolidation gain enabled by virtualization. As an example, we used polynomial functions to model a VM WCPC for different server architectures. Despite their simplicity, these models achieved strong accuracy, with a MAPE of 6.6% across representative hardware. This level of error is close to the theoretical minimum, given the inherent variability of power measurements at low utilization.

Considering dynamic resource usage in carbon accounting reveals important patterns. It quantifies energy efficiency differences between instance types providing the same service under different guarantees (via the CINERGY ratio) and demonstrates that current carbon accounting tools can significantly underestimate CO<sub>2</sub> emissions for certain VM profiles.

Beyond its theoretical contributions, this work has practical implications. First, it can be seamlessly integrated into cloud architectures to enhance existing carbon tracking tools. Second, it provides a standardized baseline for evaluating and communicating the energy efficiency benefits of cloud providers' software stacks, particularly in consolidation scenarios. Future work includes integrating these principles into the PowerAPI framework [26], enabling unified access to different types of power profiling tools. We also plan to extend our approach to account for the power consumption of additional physical resources, such as disks and network.

## SOFTWARE ARTEFACTS

The Python toolkit used to generate WCPC models is available in the first repository.<sup>3</sup> To ensure reproducibility, the evaluation code (linked to model generation) is also provided as notebooks. The CINERGY prototype, which exposes the model to users, is available in a second repository.<sup>4</sup>

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## REFERENCES

- [1] P. Jacquet, C. Coti, M. D. De Assunção, and R. Rouvoy, “Cinergy: Reasoning over the worst case power consumption of cloud virtual machines,” in *2025 IEEE 25th International Symposium on Cluster, Cloud and Internet Computing (CCGrid)*, pp. 586–589, IEEE, 2025.
- [2] International Energy Agency, “Data centres and data transmission networks,” 2021. Available at <https://www.iea.org/energy-system/buildings/data-centres-and-data-transmission-networks#programmes>
- [3] GCP, “Carbon footprint reporting methodology,” 2024. Available at <https://cloud.google.com/carbon-footprint/docs/methodology>
- [4] Microsoft, “Microsoft cloud for sustainability api calculation methodology,” 2024. Available at <https://learn.microsoft.com/en-us/industry/sustainability/api-calculation-method>
- [5] AWS, “Understanding your carbon emission estimations,” 2024. Available at <https://docs.aws.amazon.com/awsaccountbilling/latest/aboutv2/ccft-estimation.html>
- [6] GeSI, “Ict sector guidance built on the ghg protocol product life cycle accounting and reporting standard,” 2017. Available at <https://ghgprotocol.org/sites/default/files/2023-03/GHGP-ICTSG%20-%20ALL%20Chapters.pdf>
- [7] Boavizta, “Understanding the results of cloud providers' carbon calculator,” 2023. Available at <https://boavizta.org/en/blog/calculettes-carbone-clouds-providers>
- [8] S. Georgiou, S. Rizou, and D. Spinellis, “Software development lifecycle for energy efficiency: Techniques and tools,” *ACM Comput. Surv.*, vol. 52, Aug. 2019.
- [9] F. Peñañherra, J. Hobohm, and K. Szczepaniak, *LCA of Energy and Material Demands in Professional Data Centers: Case Study of a Server*, pp. 79–88. Cham: Springer International Publishing, 2019.
- [10] B. Li, R. Basu Roy, D. Wang, S. Samsi, V. Gadepally, and D. Tiwari, “Toward sustainable hpc: Carbon footprint estimation and environmental implications of hpc systems,” in *Proceedings of the International Conference for High Performance Computing, Networking, Storage and Analysis, SC '23*, (New York, NY, USA), Association for Computing Machinery, 2023.
- [11] C. Möbius, W. Dargie, and A. Schill, “Power consumption estimation models for processors, virtual machines, and servers,” *IEEE Transactions on Parallel and Distributed Systems*, vol. 25, no. 6, pp. 1600–1614, 2013.
- [12] Intel, “Intelligent platform management interface specification,” 2013. Available at <https://www.intel.fr/content/www/fr/fr/products/docs/servers/ipmi/ipmi-second-gen-interface-spec-v2-rev1-1.html>
- [13] M. Agusti, E. Caron, B. Fichel, L. Lefèvre, O. Nicol, and A.-C. Orgerie, “PowerHeat: A non-intrusive approach for estimating the power consumption of bare metal water-cooled servers,” in *GreenCom 2024 - 20th IEEE International conference on Green Computing and Communications*, pp. 1–7, Aug. 2024.
- [14] H. David, E. Gorbatov, U. R. Hanebutte, R. Khanna, and C. Le, “Rapl: Memory power estimation and capping,” in *Proceedings of the 16th ACM/IEEE international symposium on Low power electronics and design*, pp. 189–194, 2010.
- [15] M. Jay, V. Ostapenko, L. Lefèvre, D. Trystram, A.-C. Orgerie, and B. Fichel, “An experimental comparison of software-based power meters: focus on cpu and gpu,” in *2023 IEEE/ACM 23rd International Symposium on Cluster, Cloud and Internet Computing (CCGrid)*, 2023.
- [16] A. Lacoste, A. Luccioni, V. Schmidt, and T. Dandres, “Quantifying the carbon emissions of machine learning,” *arXiv preprint arXiv:1910.09700*, 2019.
- [17] L. Lannelongue, J. Grealey, and M. Inouye, “Green algorithms: quantifying the carbon footprint of computation,” *Advanced science*, vol. 8, no. 12, p. 2100707, 2021.
- [18] L. F. W. Anthony, B. Kanding, and R. Selvan, “Carbontracker: Tracking and predicting the carbon footprint of training deep learning models,” *arXiv preprint arXiv:2007.03051*, 2020.
- [19] B. Courty, V. Schmidt, S. Luccioni, Goyal-Kamal, MarionCoutarel, B. Feld, J. Lecourt, LiamConnell, A. Saboni, Inimaz, supatomic, M. Léval, L. Blanche, A. Cruveiller, ouminasara, F. Zhao, A. Joshi, A. Bogroff, H. de Lavoreille, N. Laskaris, E. Abati, D. Blank, Z. Wang, A. Catovic, M. Alencon, Michał Stechly, C. Bauer, L. O. N. de Araújo, JPW, and MinervaBooks, “mlco2/codecarbon: v2.4.1,” May 2024.
- [20] W. Lin, W. Wu, H. Wang, J. Z. Wang, and C.-H. Hsu, “Experimental and quantitative analysis of server power model for cloud data centers,” *Future Generation Computer Systems*, vol. 86, pp. 940–950, 2018.

<sup>3</sup><https://github.com/jacquetpi/cinergy-models>

<sup>4</sup><https://github.com/jacquetpi/cinergy-prototype>



- [21] W. Wu, W. Lin, L. He, G. Wu, and C.-H. Hsu, "A power consumption model for cloud servers based on elman neural network," *IEEE Transactions on Cloud Computing*, vol. 9, no. 4, pp. 1268–1277, 2019.
- [22] Scaphandre, "Scaphandre documentation," 2024. Available at <https://hubblo-org.github.io/scaphandre-documentation/>.
- [23] A. Noureddine, "Powerjoular and joularjx: Multi-platform software power monitoring tools," in *2022 18th International Conference on Intelligent Environments (IE)*, pp. 1–4, 2022.
- [24] G. Fieni, R. Rouvoy, and L. Seinturier, "Selfwatts: On-the-fly selection of performance events to optimize software-defined power meters," in *CCGRID*, pp. 324–333, IEEE, 2021.
- [25] M. Colmant, R. Rouvoy, M. Kurpicz, A. Sobe, P. Felber, and L. Seinturier, "The next 700 cpu power models," *Journal of Systems and Software*, vol. 144, pp. 382–396, 2018.
- [26] G. Fieni, D. R. Acero, P. Rust, and R. Rouvoy, "Powerapi: A python framework for building software-defined power meters," *Journal of Open Source Software*, vol. 9, no. 98, p. 6670, 2024.
- [27] E. Cadorel and D. Saingre, "A protocol to assess the accuracy of process-level power models," in *2024 IEEE International Conference on Cluster Computing (CLUSTER)*, 2024.
- [28] O. Hadary, L. Marshall, I. Menache, A. Pan, E. E. Greeff, D. Dion, S. Dorminey, S. Joshi, Y. Chen, M. Russinovich, and T. Moscibroda, "Protean: VM allocation service at scale," in *14th USENIX Symposium on Operating Systems Design and Implementation, OSDI'20*, pp. 845–861, USENIX Association, Nov. 2020.
- [29] A. Kansal, F. Zhao, J. Liu, N. Kothari, and A. A. Bhattacharya, "Virtual machine power metering and provisioning," in *Proceedings of the 1st ACM symposium on Cloud computing*, pp. 39–50, 2010.
- [30] A. E. Husain Bohra and V. Chaudhary, "Vmeter: Power modelling for virtualized clouds," in *2010 IEEE International Symposium on Parallel & Distributed Processing, Workshops and Phd Forum (IPDPSW)*, 2010.
- [31] Tads, "Energy consumption measurements in docker," in *2017 IEEE 41st Annual Computer Software and Applications Conference (COMPSAC)*, vol. 2, pp. 272–273, 2017.
- [32] M. Colmant, M. Kurpicz, P. Felber, L. Huertas, R. Rouvoy, and A. Sobe, "Process-level power estimation in vm-based systems," in *Proceedings of the Tenth European Conference on Computer Systems*, pp. 1–14, 2015.
- [33] OVHcloud, "Méthodologie du tracker d'impact environnemental d'ovhcloud," 2025. In French: [https://corporate.ovhcloud.com/sites/default/files/2025-07/methodo\\_impacttracker\\_fr.pdf](https://corporate.ovhcloud.com/sites/default/files/2025-07/methodo_impacttracker_fr.pdf).
- [34] L. A. Barroso and U. Hölzle, "The case for energy-proportional computing," *Computer*, vol. 40, no. 12, pp. 33–37, 2007.
- [35] P. Jacquet, T. Ledoux, and R. Rouvoy, "Cloudfactory: An open toolkit to generate production-like workloads for cloud infrastructures," in *11th IEEE International Conference on Cloud Engineering, IC2E'23*, 2023.
- [36] P. Jacquet, T. Ledoux, and R. Rouvoy, "SweetspotVM: Oversubscribing CPU without Sacrificing VM Performance," in *2024 IEEE 24th International Symposium on Cluster, Cloud and Internet Computing (CCGrid)*, pp. 148–157, IEEE Computer Society, May 2024.
- [37] P. Jacquet and A. Luxey-Bitri, "The case for time-shared computing resources," in *11th Workshop on Computing within Limits (LIMITS)*, 2025.
- [38] E. Cortez, A. Bonde, A. Muzio, M. Russinovich, M. Fontoura, and R. Bianchini, "Resource central: Understanding and predicting workloads for improved resource management in large cloud platforms," in *26th Symposium on Operating Systems Principles, SOSP'17*, p. 153–167, ACM, 2017.
- [39] N. Bashir, N. Deng, K. Rzaqca, D. Irwin, S. Kodak, and R. Inagal, "Take it to the limit: Peak prediction-driven resource over-commitment in datacenters," in *16th European Conference on Computer Systems, EuroSys'21*, p. 556–573, ACM, 2021.
- [40] P. Jacquet, T. Ledoux, and R. Rouvoy, "SCROOGEVM: boosting cloud resource utilization with dynamic oversubscription," *IEEE Trans. Sustain. Comput.*, vol. 9, no. 5, pp. 754–765, 2024.
- [41] Kepler, "Kubernetes efficient power level exporter (kepler)," 2024. Available at <https://sustainable-computing.io/>.
- [42] G. Fieni, R. Rouvoy, and L. Seinturier, "Smartwatts: Self-calibrating software-defined power meter for containers," in *2020 20th IEEE/ACM International Symposium on Cluster, Cloud and Internet Computing (CCGRID)*, pp. 479–488, IEEE, 2020.
- [43] P. Jacquet, T. Ledoux, and R. Rouvoy, "Slackvm: Packing virtual machines in oversubscribed cloud infrastructures," in *CLUSTER*, pp. 190–201, IEEE, 2024.
- [44] Y. Gan, Y. Zhang, D. Cheng, A. Shetty, P. Rath, N. Katarki, A. Bruno, J. Hu, B. Ritchken, B. Jackson, K. Hu, M. Pancholi, Y. He, B. Clancy, C. Colen, F. Wen, C. Leung, S. Wang, L. Zaruvisky, M. Espinosa, R. Lin, Z. Liu, J. Padilla, and C. Delimitrou, "An Open-Source Benchmark Suite for Microservices and Their Hardware-Software Implications for Cloud & Edge Systems," in *ASPLOS*, pp. 3–18, ACM, 2019.
- [45] Z. Ourmani, M. C. Belgaid, R. Rouvoy, P. Rust, J. Penhoat, and L. Seinturier, "Taming energy consumption variations in systems benchmarking," in *Proceedings of the ACM/SPEC International Conference on Performance Engineering, ICPE '20*, p. 36–47, ACM, 2020.
- [46] Google Cloud Platform, "Preemptible vm instances," 2020. Available at <https://cloud.google.com/compute/docs/instances/preemptible>.
- [47] Y. Wang, K. Arya, M. Kogias, M. Vanga, A. Bhandari, N. J. Yadwadkar, S. Sen, S. Elnikety, C. Kozyrakis, and R. Bianchini, "Smartharvest: Harvesting idle cpus safely and efficiently in the cloud," in *16th European Conference on Computer Systems, EuroSys'21*, ACM, 2021.
- [48] G. Fieni, R. Rouvoy, and L. Seinturier, "x pue: Extending power usage effectiveness metrics for cloud infrastructures," *IEEE Transactions on Sustainable Computing*, 2025.



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